

Credit Quantity, Credit Prices, and Future Output

A model of heterogeneous financing access – Track B

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<https://github.com/amchavezu/ai-project>

When does the **quantity** of bank credit contain more information about future production than the **price** of credit?

- ▶ Track B: formulate the firm model missing from my undergraduate thesis.
- ▶ Preliminary estimates suggest different credit indicators may be useful for non-tradable and tradable activity.
- ▶ I treat that evidence as qualitative and unfinished. The model targets the cross-sector mechanism, not exact horizons or causality.

1 · The pattern that needs a mechanism

Preliminary quantity pattern

Bank credit volume appears more informative for non-tradable activity at a relatively short horizon.

Candidate mechanism: bank-dependent firms cannot replace a missing loan.

Preliminary price pattern

A credit-price indicator may become more informative for tradable activity over a longer horizon.

Candidate mechanism: firms with alternative finance choose scale using the marginal funding cost.

These are working hypotheses from a thesis in progress, not final empirical findings.

At date zero, firm i has internal funds n_i and chooses capital k_i , bank credit b_i , and outside finance m_i :

$$\max_{k_i, b_i, m_i} \quad \beta F_i(k_i) - R_B b_i - R_M m_i$$

$$k_i = n_i + b_i + m_i, \quad 0 \leq b_i \leq \bar{b}_i, \quad m_i \geq 0.$$

- ▶ $F'_i(k) > 0$ and $F''_i(k) < 0$.
- ▶ $0 < R_B \leq R_M = R_B + \tau_i$: bank funds are cheaper but capped.
- ▶ A bank-dependent firm has $m_i = 0$; a market-access firm may have $m_i > 0$.

2 · Two margins of transmission

Let $\bar{k}_i = n_i + \bar{b}_i$ be capital after exhausting the bank line.

	Quantity-sensitive kink	Price-sensitive margin
Marginal finance	No cheap substitute for the bank line	Outside finance
Choice of capital	$k_i^* = \bar{k}_i$	$\beta F'_i(k_i^*) = R_M$
Larger bank line	Raises total capital	Replaces outside finance
Higher common spread	No local scale effect at the kink	Lowers desired capital

The comparison is local: small changes must leave the firm in the same regime.

3 · The result I expect

With $F_i(k) = a_i k - c_i k^2/2$, define

$$k_B = \frac{\beta a_i - R_B}{\beta c_i}, \quad k_M = \frac{\beta a_i - R_M}{\beta c_i}, \quad k_M \leq k_B.$$

$$k_i^* = \begin{cases} n_i, & k_B \leq n_i, \\ k_B, & n_i < k_B < \bar{k}_i, \\ \bar{k}_i, & k_M \leq \bar{k}_i \leq k_B, \\ k_M, & \bar{k}_i < k_M. \end{cases}$$

The third line is the binding quantity regime. The fourth is the market-finance regime. Strict inequalities deliver the comparative statics without a regime switch.

3 · Expected sectoral ranking

Let λ_s be the share of sector s at the quantity-sensitive kink:

$$Y_s = \lambda_s F(n + \bar{b}) + (1 - \lambda_s) F(k_M).$$

If $\lambda_N > \lambda_T$ and marginal products are positive, I expect

$$\frac{\partial Y_N}{\partial \bar{b}} > \frac{\partial Y_T}{\partial \bar{b}}, \quad \left| \frac{\partial Y_N}{\partial z} \right| < \left| \frac{\partial Y_T}{\partial z} \right|.$$

- ▶ Credit quantity signals the feasible project scale for constrained firms.
- ▶ The spread signals the marginal cost and desired project scale for firms that can substitute across funding sources.
- ▶ Adjustment speeds may affect timing, but the baseline does not derive exact forecast horizons.

Closest work	What it establishes	What remains for this project
Holmström–Tirole (1997)	Capital, monitoring, and access to direct finance	No sectoral ranking of quantity and price predictors
Gilchrist–Zakrajšek (2012)	Credit spreads predict aggregate activity	No binding bank-quantity comparison at the firm level
Müller–Verner (2024)	Sectoral credit allocation predicts different macro outcomes	No capped-bank versus marginal-market threshold

I checked the published versions, public working-paper versions, and the relevant appendices and follow-on investment paper. The contribution is a narrow synthesis tailored to the thesis, not a claim that the underlying financial-friction mechanisms are new.

Plan

- ▶ Prove the piecewise optimum and sectoral rankings.
- ▶ Check every reported result with `code/verify.py`.
- ▶ Simulate behavior near regime boundaries.
- ▶ Formalize every numbered result through `AppliedModelingLib`.

Main risks

- ▶ Outside-finance access is not directly measured in the thesis data.
- ▶ Endogenous credit limits may remove the sharp local zero.
- ▶ A static model cannot identify exact forecast timing.
- ▶ The piecewise global optimum is the hardest Lean step.

If a step fails, I will keep the result conditional and report the exact open case rather than strengthen the assumptions silently.